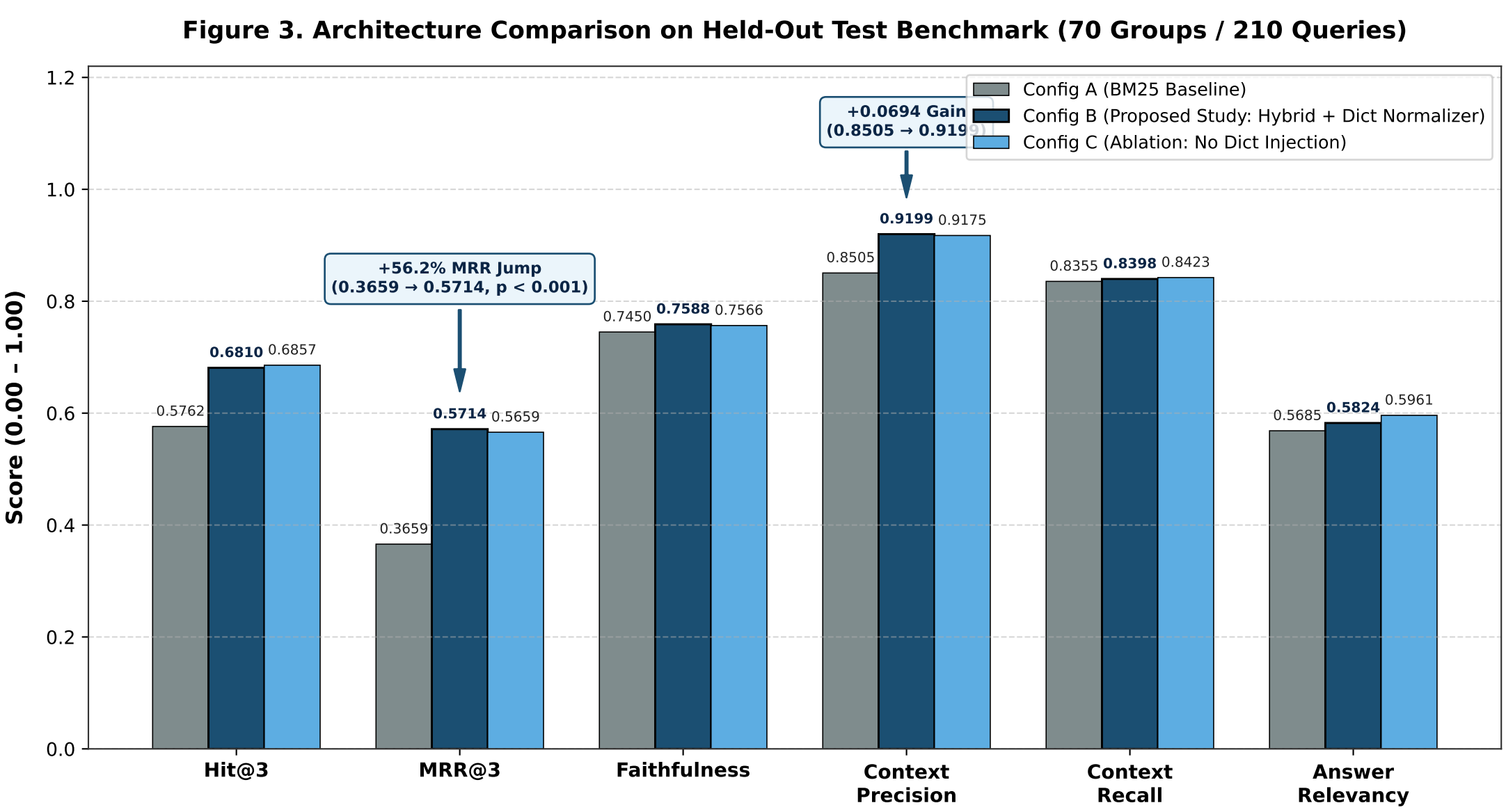
Figure 3. Architecture Comparison on Held-Out Test Benchmark (70 Groups / 210 Queries)
Config A (BM25 Baseline)
Config B (Proposed Study: Hybrid + Dict Normalizer)
Config C (Ablation: No Dict Injection)
+56.2% MRR Jump
(0.3659 → 0.5714, p < 0.001)
+0.0694 Gain
(0.8505 → 0.919
Score (0.00 – 1.00)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.5762
0.6810
0.6857
0.3659
0.5714
0.5659
0.7450
0.7588
0.7566
0.8505
0.9199
0.9175
0.8355
0.8398
0.8423
0.5685
0.5824
0.5961
Hit@3
MRR@3
Faithfulness
Context Precision
Context Recall
Answer Relevancy